

Interference Effects - Lloyd's Mirror

Laboratory Experiment #3

By

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Submitted To:

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1.) OBJECTIVE:

- To study basic interference phenomena.
- To build Lloyd's Interferometer.

2.) MATERIALS:

- | | |
|---|---|
| □ Optical Table (SG-22-2) | □ 4x Post Holders (VPH-2) |
| □ Damped Rod (45) | □ 3x 1" Diameter Adj. Mirror Mounts. (E55-532) |
| □ Adjustable Mounting Fixtures for HeNe Laser <ul style="list-style-type: none">○ Laser Mount (ULM-TILT)○ Rod Clamp (340-RC) | □ Lens Kit (E52-305) (2x DCV -25 mm; DCX +125 mm) |
| □ HeNe Laser | □ 3x Base (BA2) |
| □ Mirror $\frac{1}{4}$ Wave Al (32661 2") | □ 2x Base (B-1) |
| | □ Screw Kit, $\frac{1}{4}$ x 20 (SK-25A) |

3.) BACKGROUND:

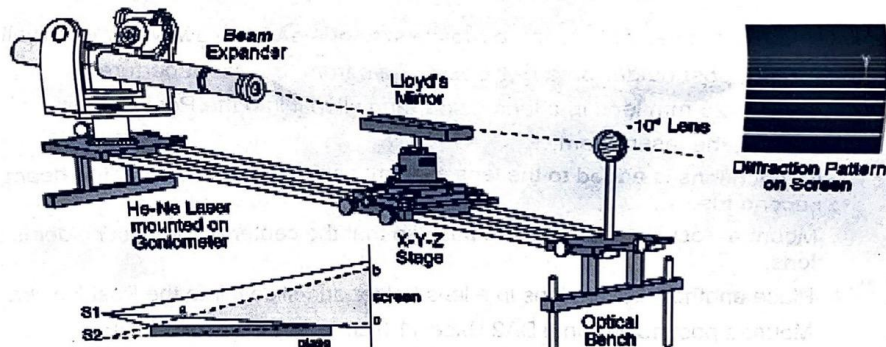


Figure 1

Lloyd's Mirror is a strip of smooth glass about six inches long, painted black on the underside. An expanded monochromatic HeNe laser beam is sent at a grazing angle to the glass so that part of the beam is reflected from the glass (causing a phase change of π) and interferes with the part of the beam that doesn't hit the glass. The interference fringes are magnified by the diverging lens and projected onto a screen.

In Fig.1, the laser beam is the source S₁. The light reflected from Lloyd's mirror appears as though it came from virtual source S₂. The overlapping region from S₁ and S₂ (abc area) is where the interference occurs, showing up as a series of fringes on the screen in the region bc.

4.) Procedure

LLOYD'S MIRROR

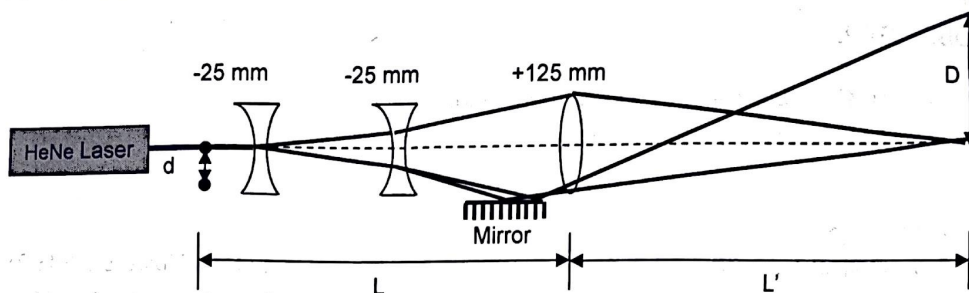
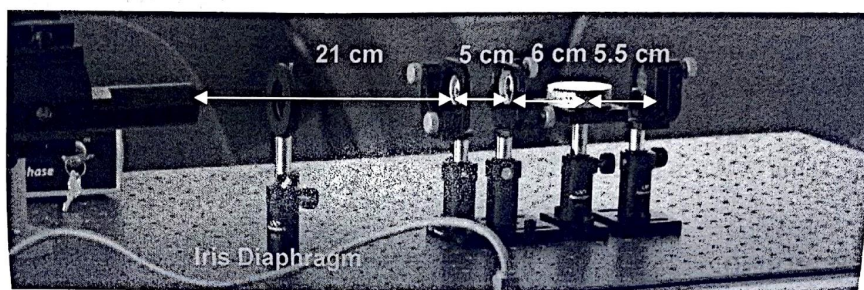


Figure 1: Experimental Setup for Lloyd's Mirror



1. Slide the optical table to the opposite side of the bench away from the wall.
2. Mount a post holder on a B-1 Base 21 cm from the laser aperture.
3. Place a -25 mm lens in a lens holder and slide it into the Post Holder.
4. Unblock the laser beam.
5. As each lens is added to the lens system, center the lens so that the beam is centered on the second Iris.
6. Mount a Post Holder on a BA2 base so that the center of the post holder is 5 cm past the first lens.
7. Place another -25 mm lens in a lens holder and slide it into the Post Holder.
8. Mount a post holder on a BA2 Base 11.5 cm past the second lens.
9. Place a +125 mm lens into a Lens Holder and slide it into the Post Holder.
10. Adjust the +125 mm lens so that the spot projected onto the wall is as small as possible.
11. Measure the distance L' between the positive lens and the wall. Record L' value in Table 1.
12. Measure the distance between the first negative lens and the positive lens. It should be 16.5 cm. Add to this distance the focal length of 0.625 cm (the focal lens of the two negative lens system). This is L , as shown in Figure 1. Record L in Table 1.
13. Mount a Post Holder between the -25 mm lens and the +125 mm lens using a B-1 Base.
14. Mount a BA2 Base to a post and slide it into the Post Holder.
15. Place the $\frac{1}{4}$ Wave mirror on the BA2 base.
16. The distance between the second negative lens and the center of the mirror should be 6 cm.

17. Remove the second iris.
18. Adjust the height of the mirror so that a second spot appears on the wall about one inch above the first spot.
19. Measure the distance D between the two spots and record it in **Table 1**.
20. A virtual source appears at a distance d from the actual source. This distance can be determined using the relationship:

$$d = \frac{D \cdot L}{L'}$$

where: D is the distance between the two spots on the wall;

L is the object distance;

L' is the image distance. Calculate and record d in **Table 1**.

21. Remove the positive lens from the setup. The light beams coming from the two sources will partially overlap forming an interference pattern on the wall (see **Fig.3**). This pattern will consist of alternate light and dark horizontal lines.



Figure 3 – Interference Pattern

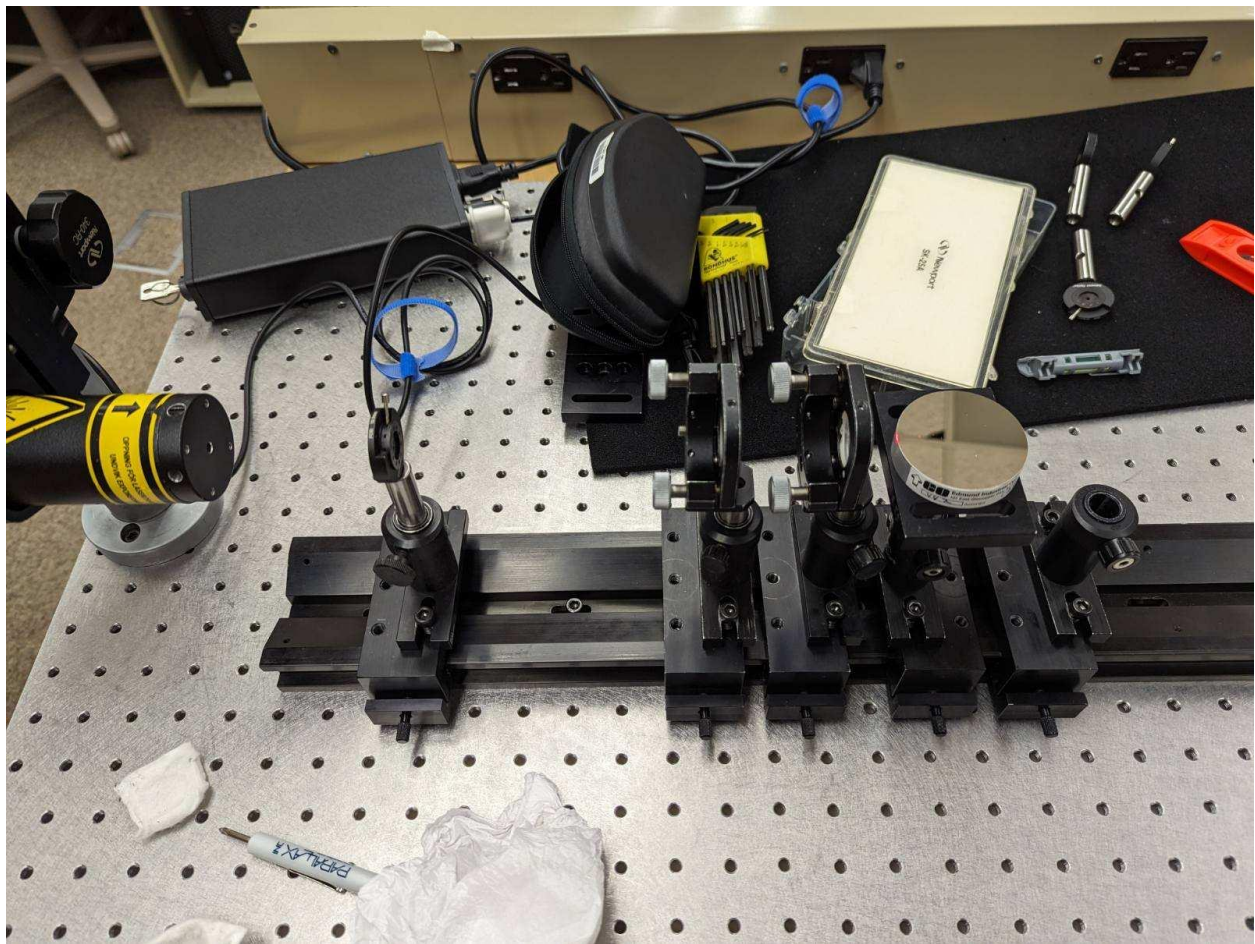
22. Using a magnifying glass, measure the distance between two consecutive *dark fringes*. To improve the accuracy, count the number of dark fringes along a (5 mm) distance. Mark on the paper two points (5 mm far from each other) using a pencil, near the center of the interference pattern. Divide the distance (5 mm) by the number of dark fringes and record this result (x) in **Table 1**.
23. Block the laser beam. Using the distance (x) between two dark fringes determine the value of $\sin(\theta)$:

$$\sin(\theta) = \frac{x}{(L + L')}$$

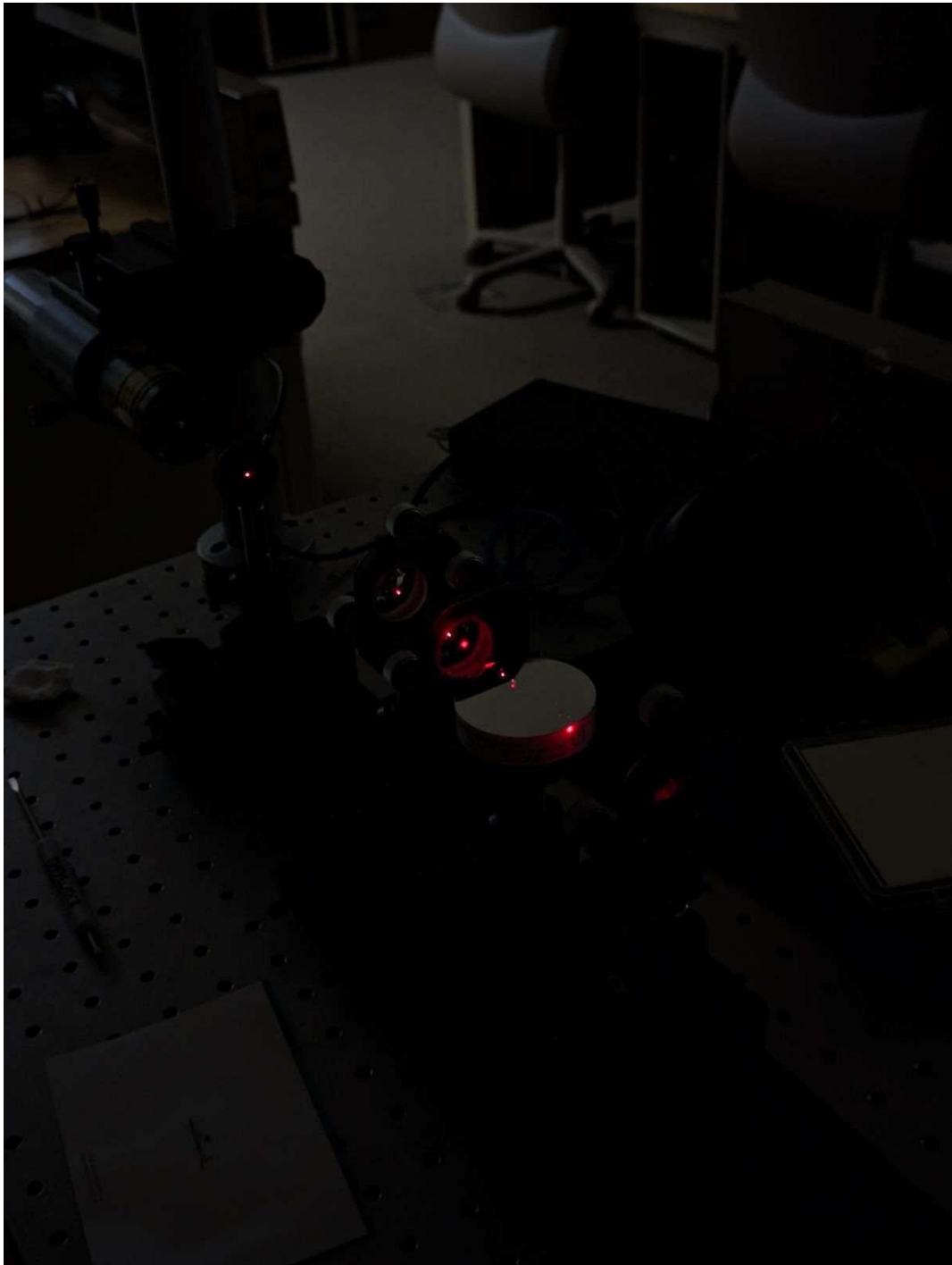
$\sin(\theta)$ is the distance between any two bright fringes or any two dark fringes on the screen divided by the distance between the screen and the source. Record it in **Table 1**.

24. Calculate the angle $\theta = \underline{\hspace{2cm}}$. Is there a difference between $\sin(\theta)$ and θ ? Explain.
25. Calculate and record the wavelength of the HeNe laser using the relationship:

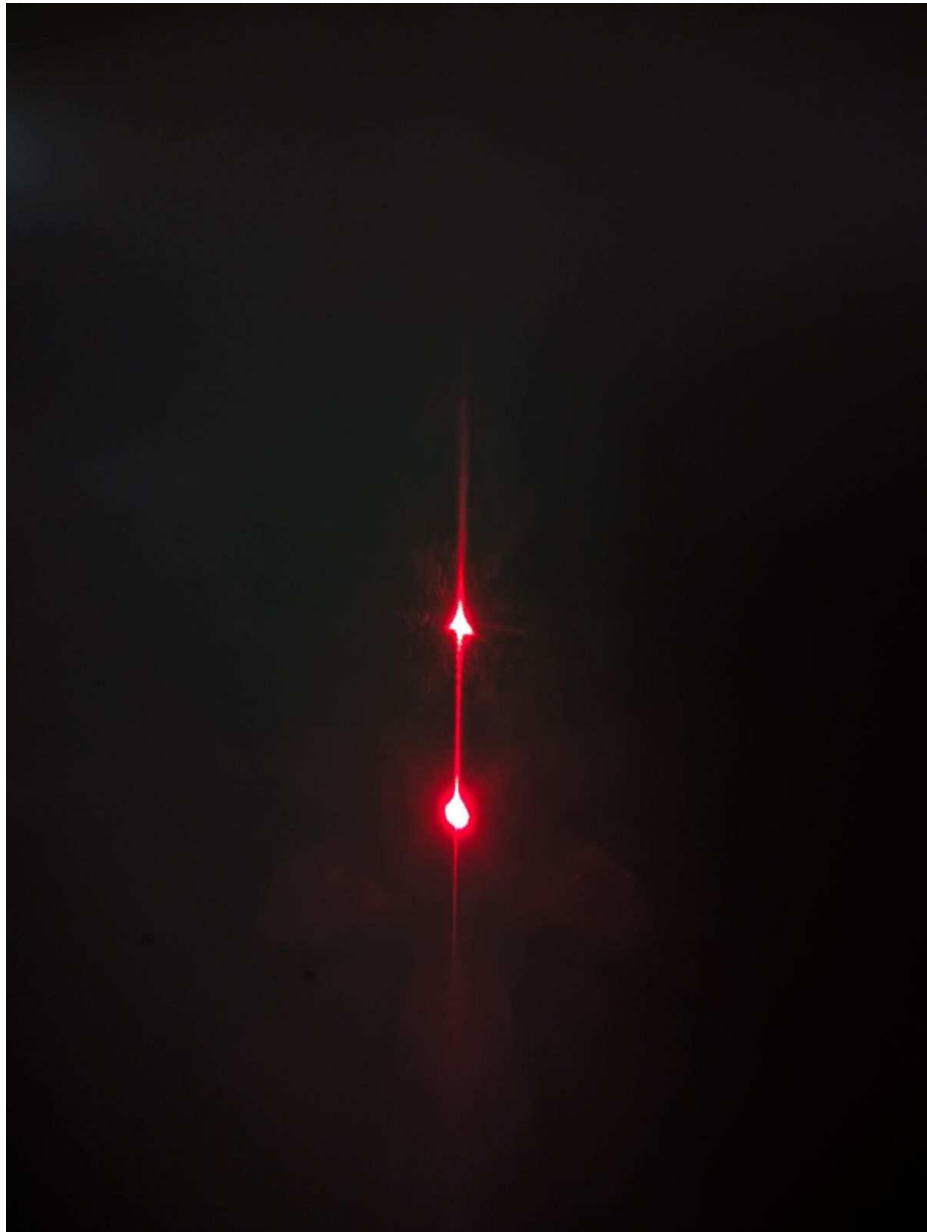
$$\lambda = d \cdot \sin(\theta)$$



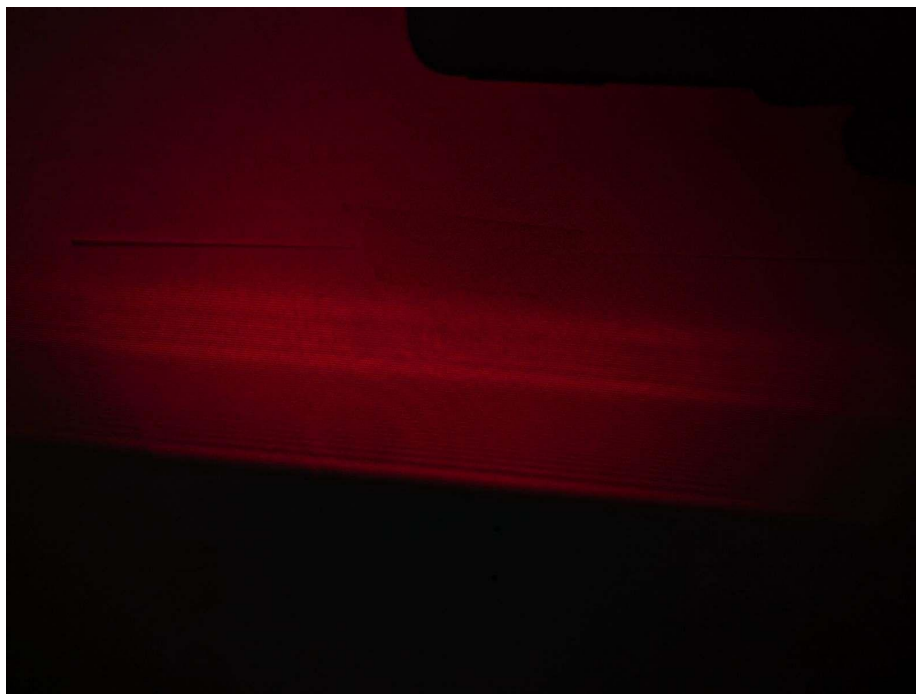
Lab setup.



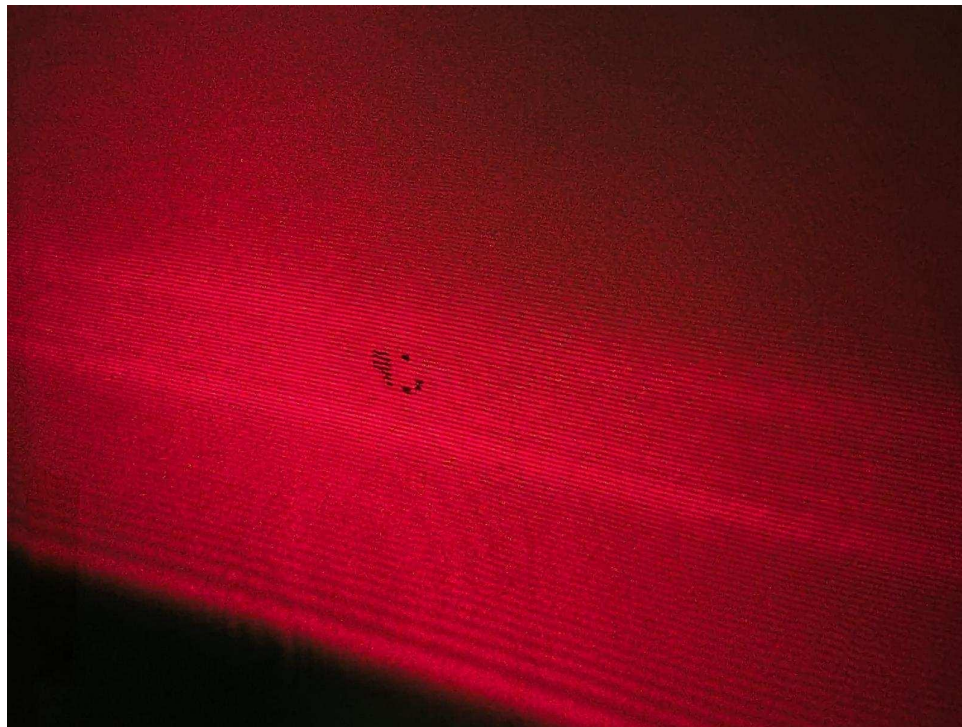
Beam light passing through optics and rejecting off the mirror.



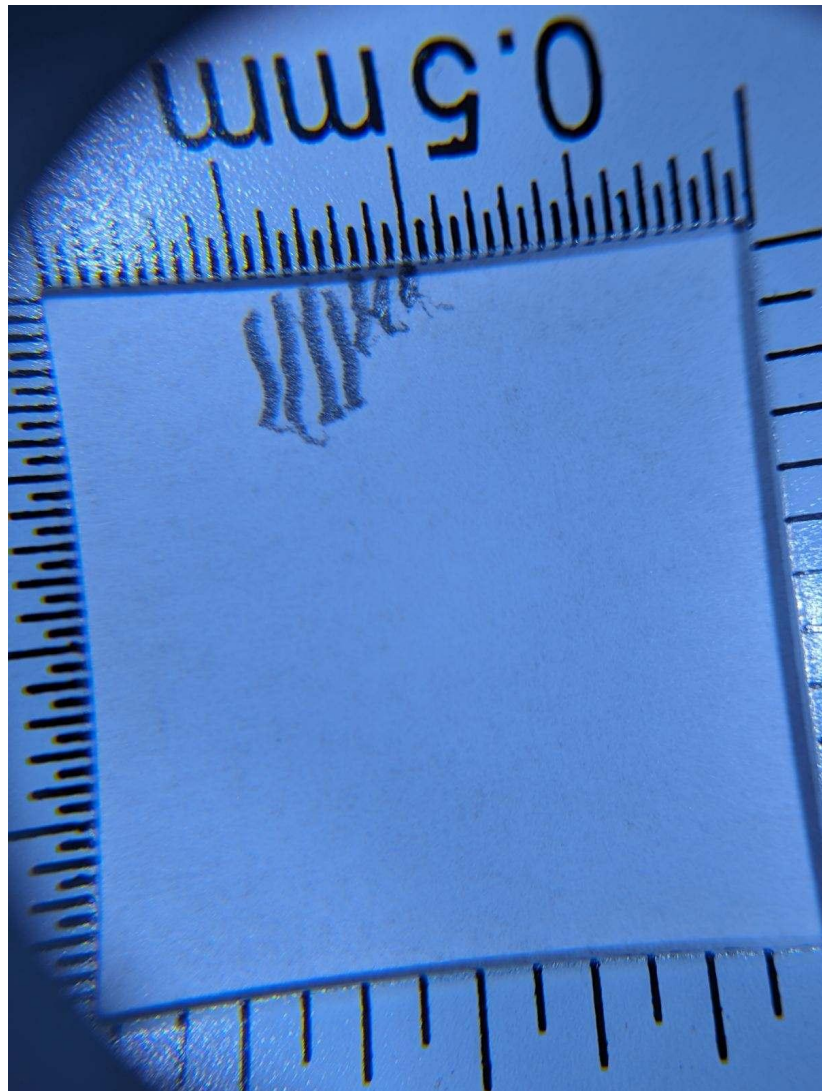
Beam light reflecting off the mirror and aligned correctly..



Interference pattern.



Marking interference pattern



Dark fringe distance measurement.

5.) Data & Observations

TABLE 1 Single Slit

L cm	L' cm	D cm	d cm	x cm	sin (θ)	λ nm
17.625	19	2.3	21.3355	0.0625	301.026	642.254

$$d = D \cdot L / L'$$

$$2.3 \text{ cm} \cdot 17.625 \text{ cm} / 19 \text{ cm} \rightarrow 21.3355 \text{ cm}$$

$$\sin(\theta) = x / (L + L')$$

$$\sin^{-1}(0.0625 / (17.625 + 19)) \rightarrow 301.026$$

$$\lambda = d \cdot \sin(\theta)$$

$$21.3355 \cdot 301.026 \rightarrow 642.254$$

The data obtained is as accurate as possible by measurement tools and the conductors of the experiment. This data shows that the interference patterns of the optics and the mirror are accurately observed. Some errors can occur during the experiment if the mirror and optics aren't aligned, measured distance or calculations are incorrect, or if the laser isn't parallel to the table.

6.) Answers to Lab Questions

24.) Calculate the angle. Is there a difference between sin (θ) and θ ? 0.1724° is theta and 301.026 degrees is sin (θ). The difference is that sin (θ) is the difference between two fringes divided by the distance between the screen and the light source, whereas theta is the angle of reflection.

7.) Conclusion

In conclusion, the laboratory experiment showed that if the optics are aligned correctly, the measurements were accurate and consistent. The experiment yielded expected results when it was properly aligned because the reflected beam from the mirror is crucial for observing the interference pattern. If the optics are not properly aligned, the beam will not reflect properly to obtain accurate measurements and will yield incorrect results. We initially had issues with optical alignment, but when we switched to a dovetail rail, it kept it more secure and

consistent. Improvements would be to ensure that user errors and equipment errors are at a minimum so that the results are as accurate as possible for future work.